
IEOR 165 Project

Due 11:59 PM PT Friday, May 1, Spring 2026

Instructions:

- The course project must be submitted on Gradescope as a PDF report with source code link included. You could upload your code to Google Drive, Github, etc, and add the link in your PDF report.
 - You are allowed to work in groups with up to 5 students total.
 - The project will be graded based on the correctness of implementing machine learning models.
 - You may use any programming language, software, library and package.
 - Please submit your project on time. For late submissions, there is a 10% point deduction per late day.
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Introduction:

In the house sales market, the price may depend on several factors including house area, number of bedrooms and bathrooms, furnishings, nearness to mainroad, etc. This case study aims to study the effect on the price of a house by such factors. We will use the dataset "house_sales.csv". Harrison, D. and Rubinfeld, D.L. (1978)

Hedonic prices and the demand for clean air. J. Environ. Economics and Management 5, 81-102. Belsley

D.A., Kuh, E. and Welsch, R.E. (1980) Regression Diagnostics. Identifying Influential Data and Sources of Collinearity. New York: Wiley.

Project Tasks:

Firstly, split the dataset to training and testing dataset. Use the training dataset to fit three models and predict the house price. The three models are given as follows.

- Ordinary least squares
- Regularized least regression (You may choose one model from ridge or LASSO)
- Nadaraya-Watson kernel regression (Select one feature for kernel regression).

For the second model, you need to determine the parameter regularization multiplier λ . For the third model, you need to determine the parameter bandwidth h . Use cross-validation on the training dataset to determine the parameter value for these two models. For these two models, you need to

- include plots of tuning parameters versus cross-validation error.

For all the three models, you need to

- present each of your models with formula clearly, i.e., a clear function $\hat{y} = f(x)$ which calculates the predicted $\hat{y}$ given data x
- test your models on the testing dataset and report the mean square error.